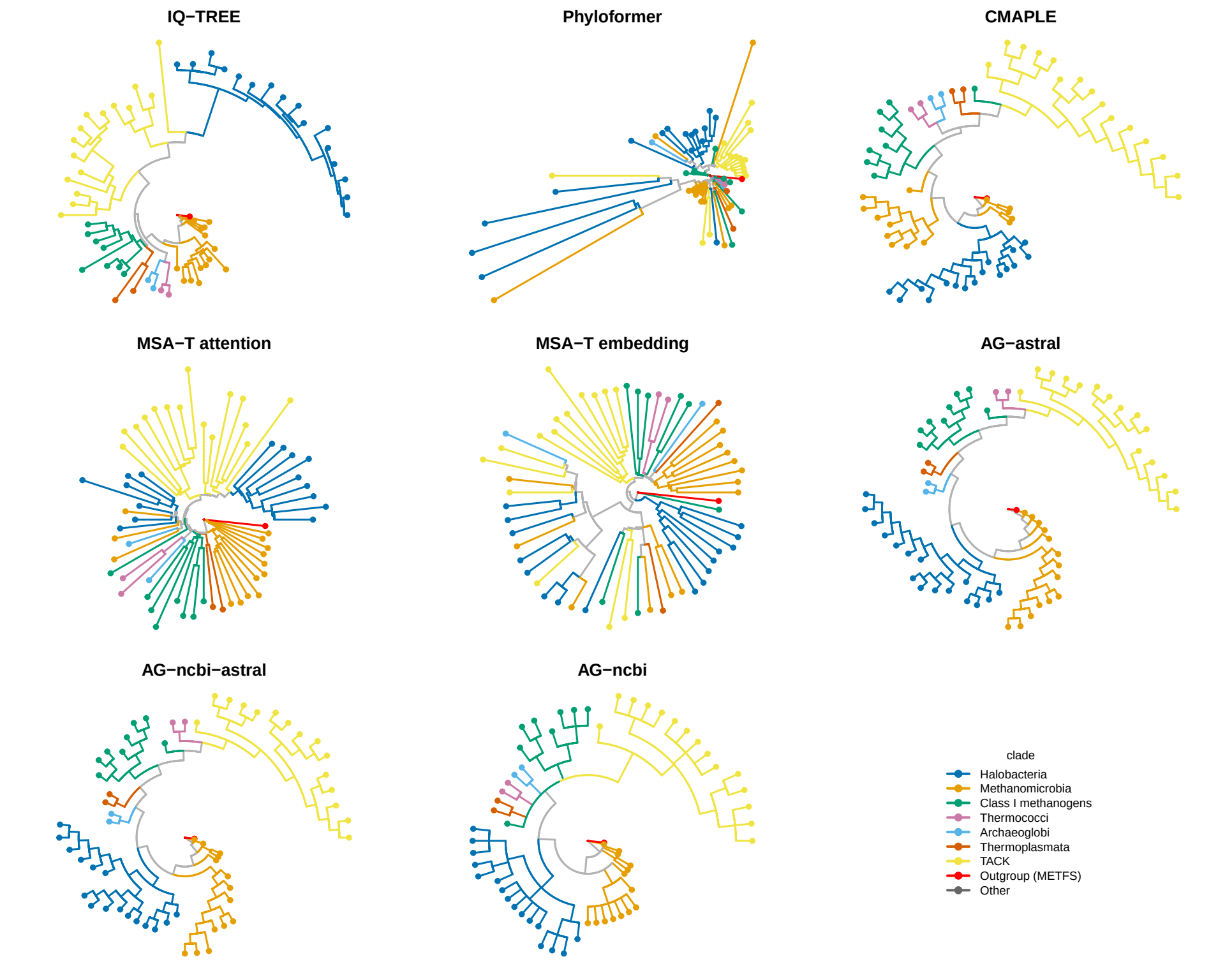
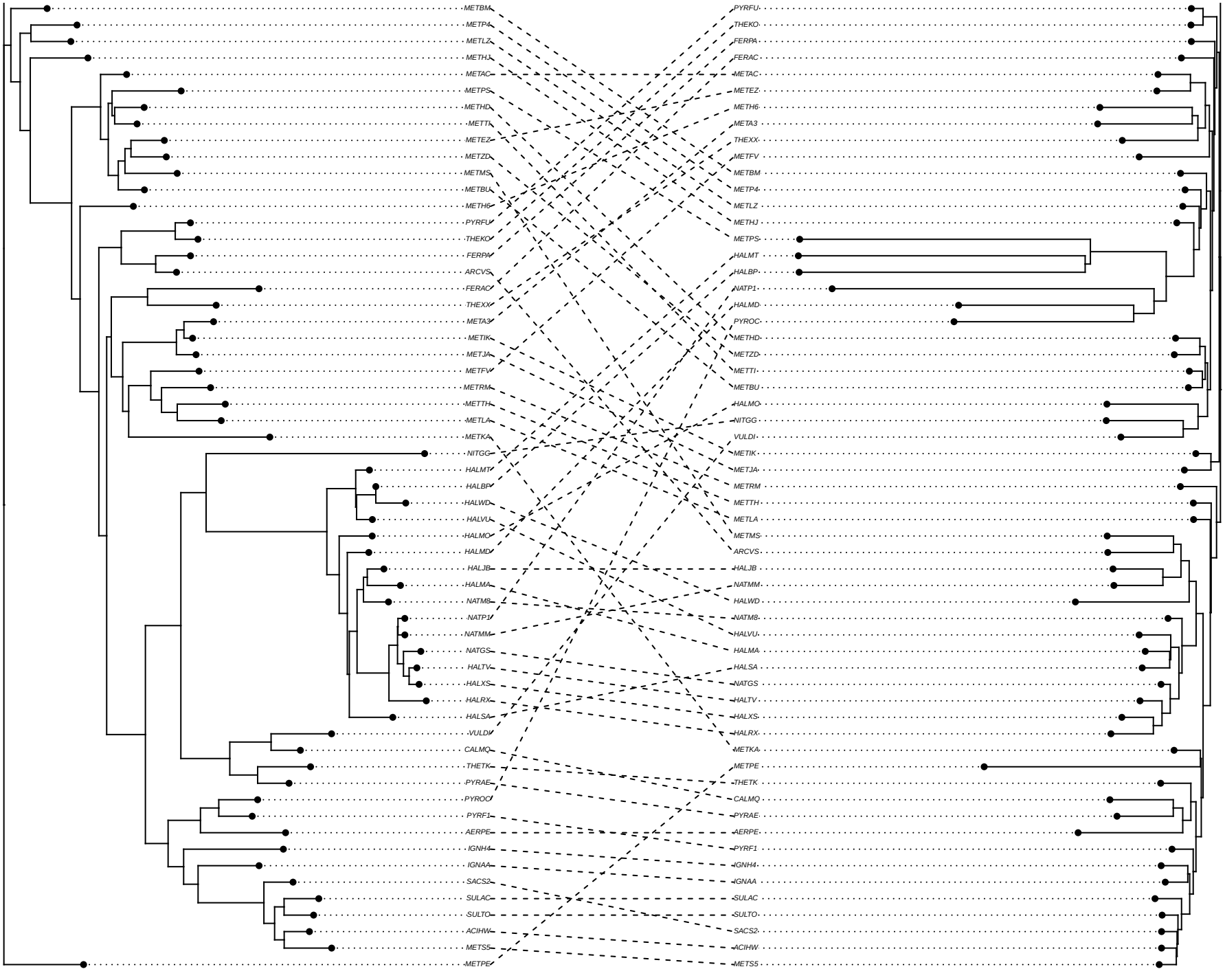
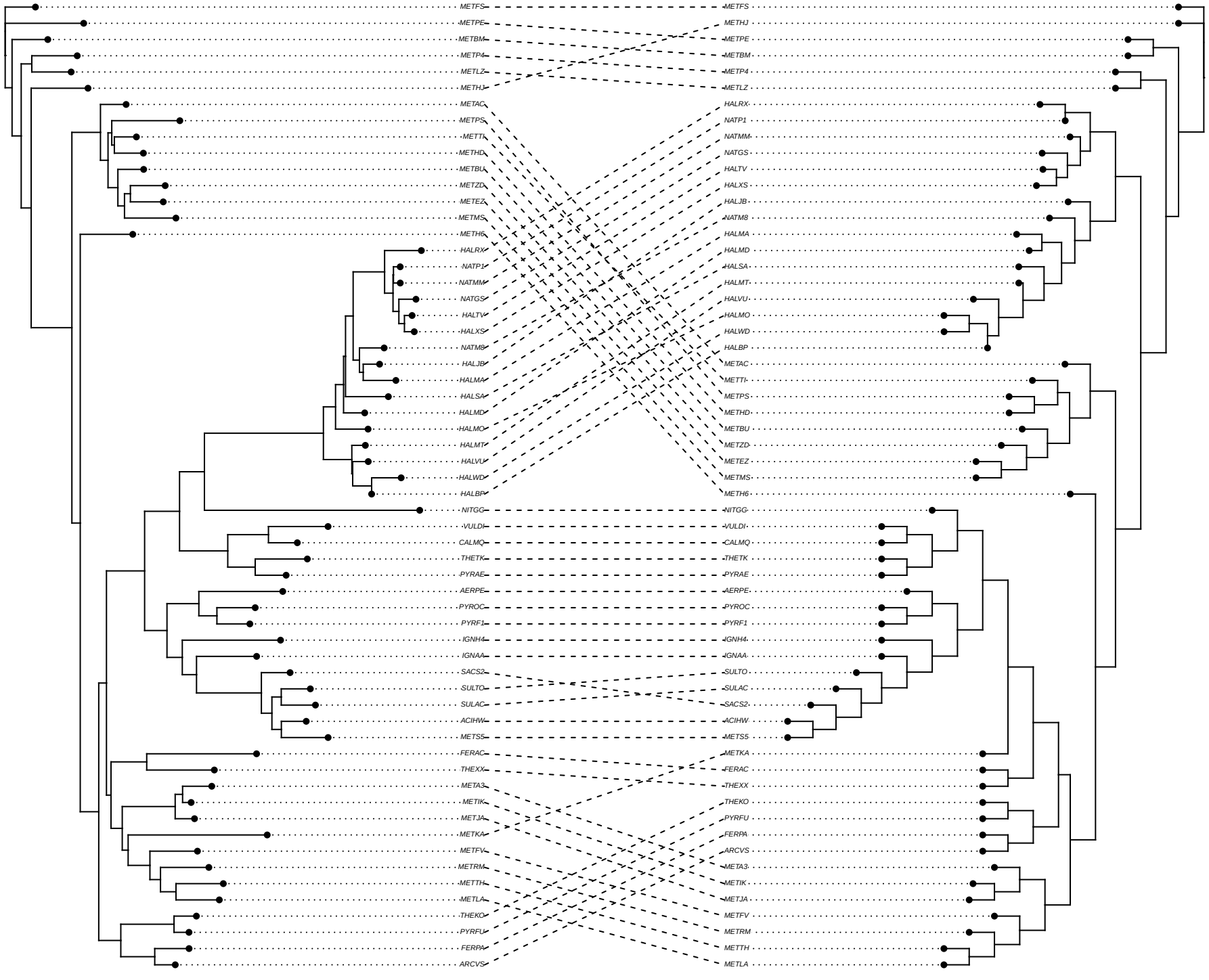
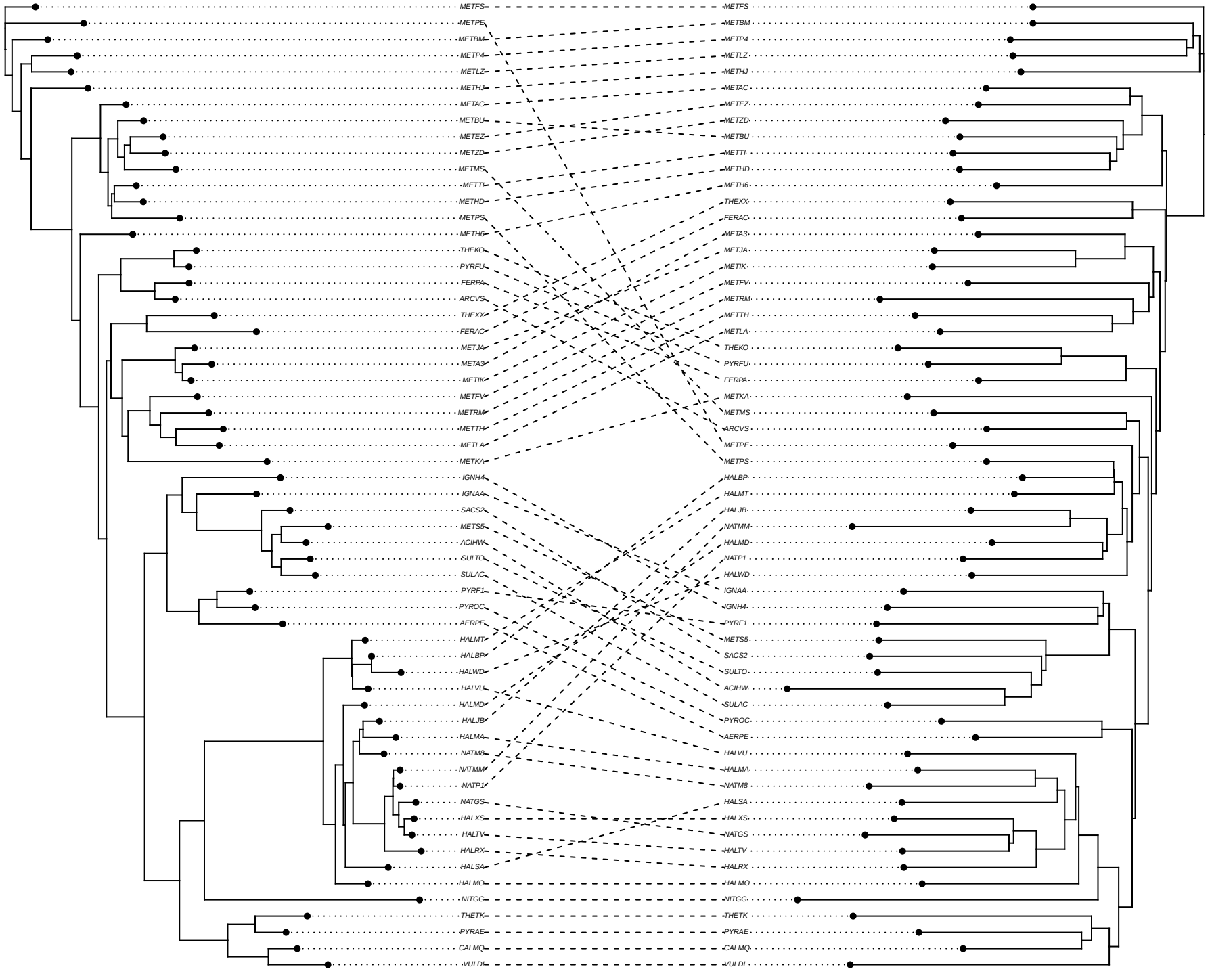

IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
AG−astral
AG−ncbi−astral
AG−ncbi
clade
Halobacteria
Methanomicrobia
Class I methanogens
Thermococci
Archaeoglobi
Thermoplasmata
TACK
Outgroup (METFS)
Other

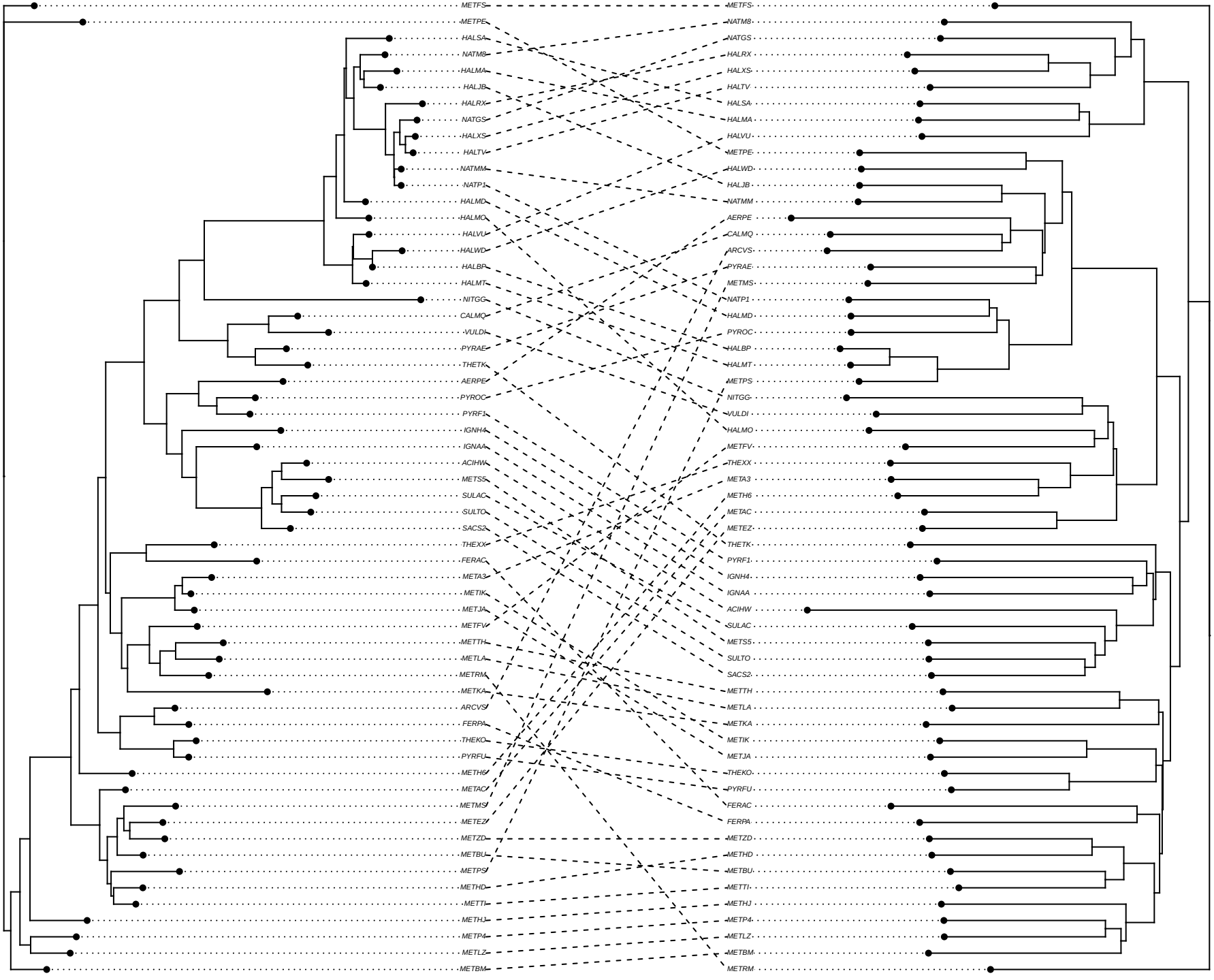

IQ−TREE vs Phylolformer
METBM
METP4
METLZ
METHJ
METAC
METPS
METHD
METTI
METEZ
METZD
METMS
METBU
METH6
PYRFU
THEKO
FERPA
ARCVS
FERAC
THEXX
META3
METIK
METJA
METFV
METRM
METTH
METLA
METKA
NITGG
HALMT
HALBP
HALWD
HALVU
HALMO
HALMD
HALJB
HALMA
NATM8
NATP1
NATMM
NATGS
HALTV
HALXS
HALRX
HALSA
VULDI
CALMQ
THETK
PYRAE
PYROC
PYRF1
AERPE
IGNH4
IGNAA
SACS2
SULAC
SULTO
ACIHW
METS5
METPE
PYRFU
THEKO
FERPA
FERAC
METAC
METEZ
METH6
META3
THEXX
METFV
METBM
METP4
METLZ
METHJ
METPS
HALMT
HALBP
NATP1
HALMD
PYROC
METHD
METZD
METTI
METBU
HALMO
NITGG
VULDI
METIK
METJA
METRM
METTH
METLA
METMS
ARCVS
HALJB
NATMM
HALWD
NATM8
HALVU
HALMA
HALSA
NATGS
HALTV
HALXS
HALRX
METKA
METPE
THETK
CALMQ
PYRAE
AERPE
PYRF1
IGNH4
IGNAA
SULAC
SULTO
SACS2
ACIHW
METS5

# IQ-TREE vs CMAPLE

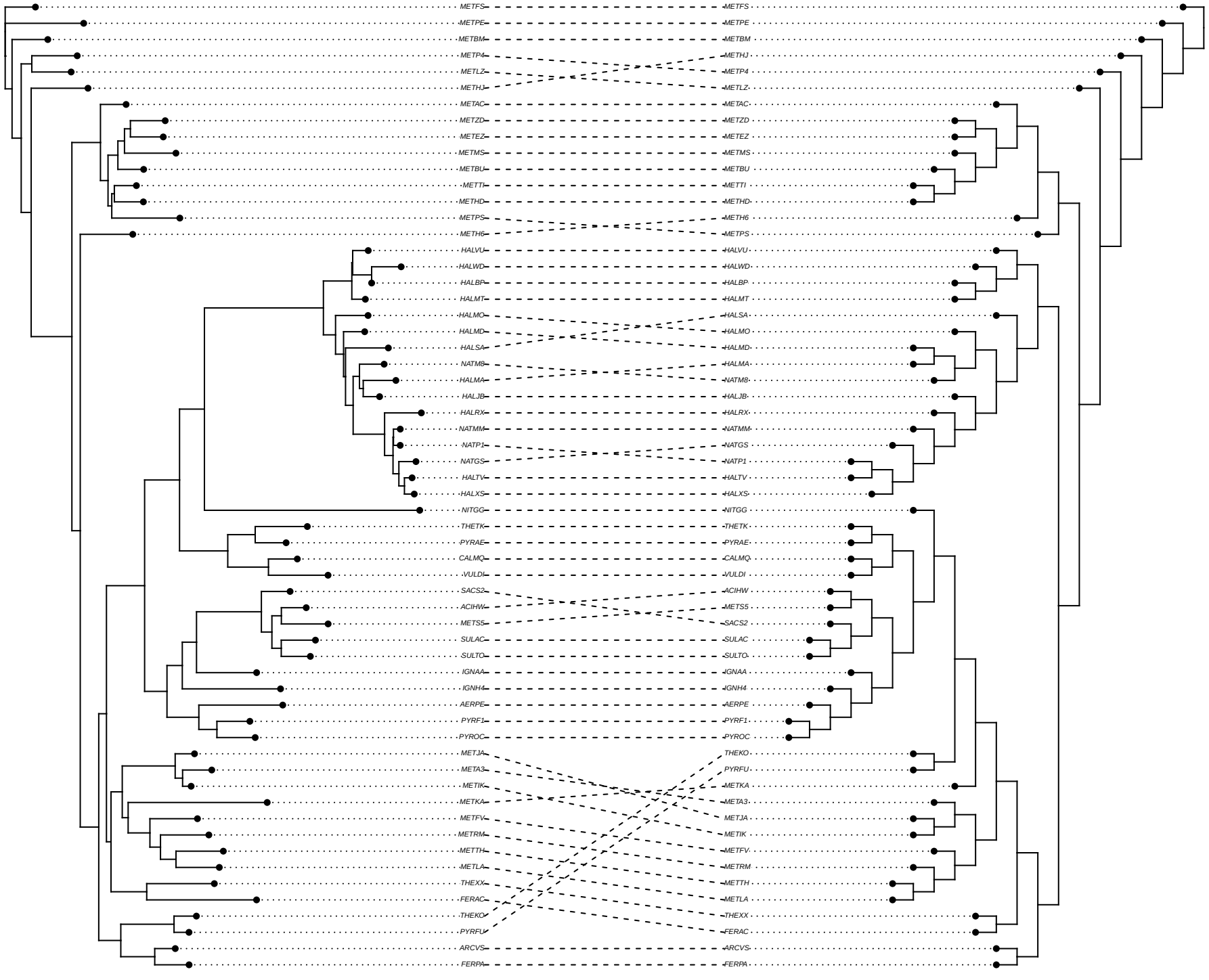



IQ−TREE vs MSA−Pattention
METFS
METPE
METBM
METP4
METLZ
METHJ
METAC
METBU
METEZ
METZD
METMS
METTI
METHD
METPS
METH6
THEKO
PYRFU
FERPA
ARCVS
THEXX
FERAC
METJA
META3
METIK
METFV
METRM
METTH
METLA
METKA
IGNH4
IGNAA
SACS2
METS5
ACIHW
SULTO
SULAC
PYRF1
PYROC
AERPE
HALMT
HALBP
HALWD
HALVU
HALMD
HALJB
HALMA
NATM8
NATMM
NATP1
NATGS
HALXS
HALTV
HALRX
HALSA
HALMO
NITGG
THETK
PYRAE
CALMQ
VULDI
METFS
METBM
METP4
METLZ
METHJ
METAC
METEZ
METZD
METBU
METTI
METHD
METH6
THEXX
FERAC
META3
METJA
METIK
METFV
METRM
METTH
METLA
THEKO
PYRFU
FERPA
METKA
METMS
ARCVS
METPE
METPS
HALBP
HALMT
HALJB
NATMM
HALMD
NATP1
HALWD
IGNAA
IGNH4
PYRF1
METS5
SACS2
SULTO
ACIHW
SULAC
PYROC
AERPE
HALVU
HALMA
NATM8
HALSA
HALXS
NATGS
HALTV
HALRX
HALMO
NITGG
THETK
PYRAE
CALMQ
VULDI

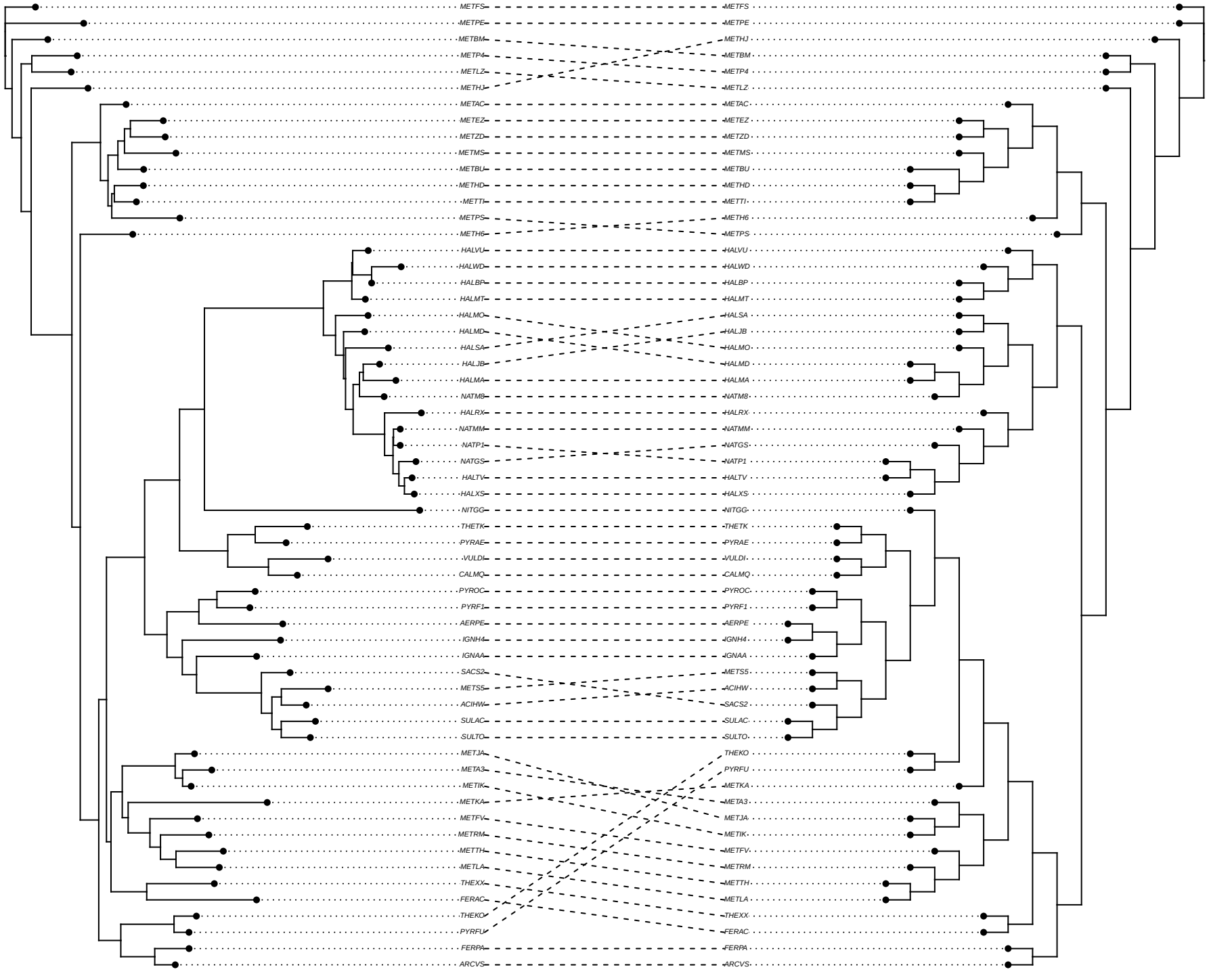
# IQ-TREE vs MSA-Transformer embedding

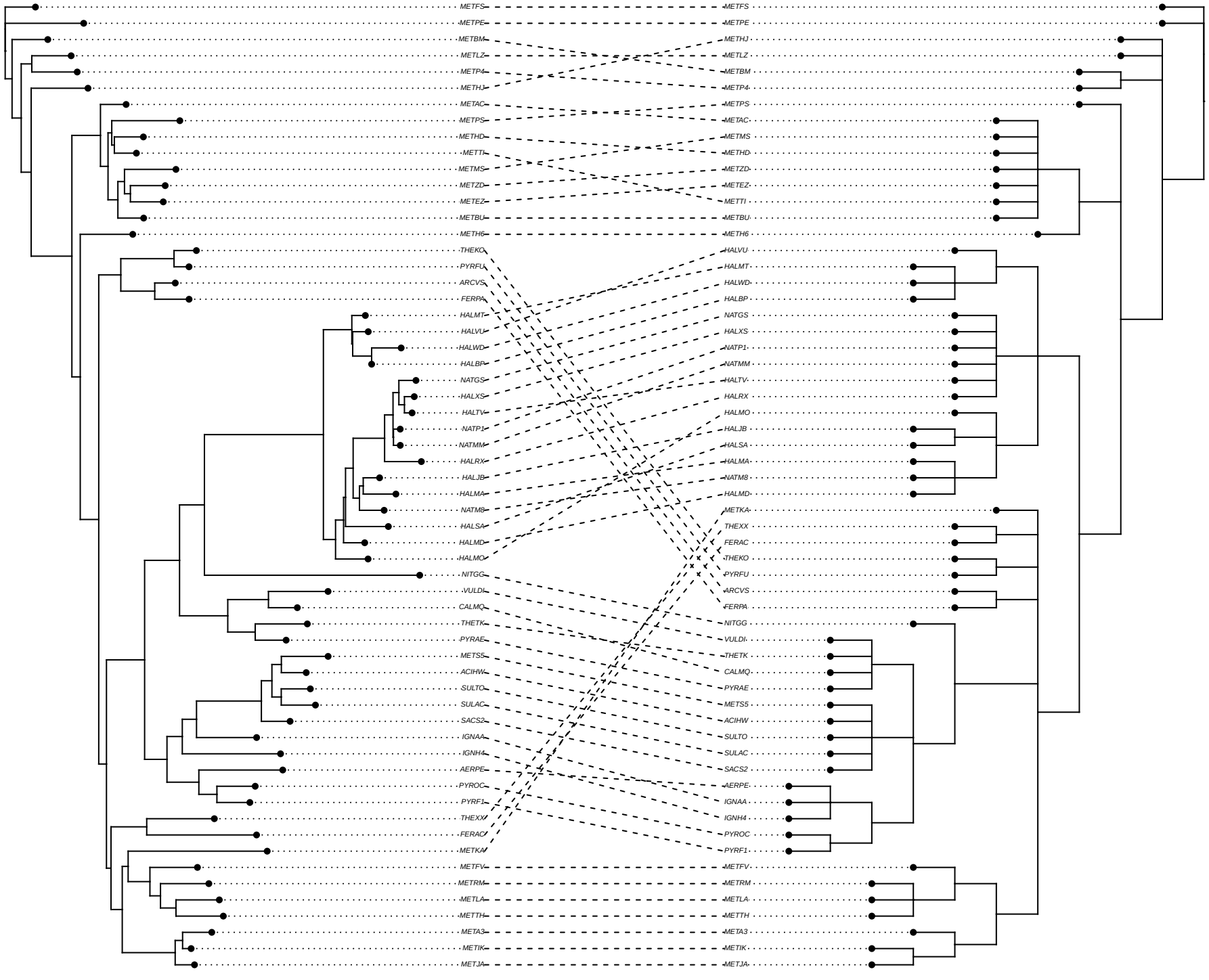


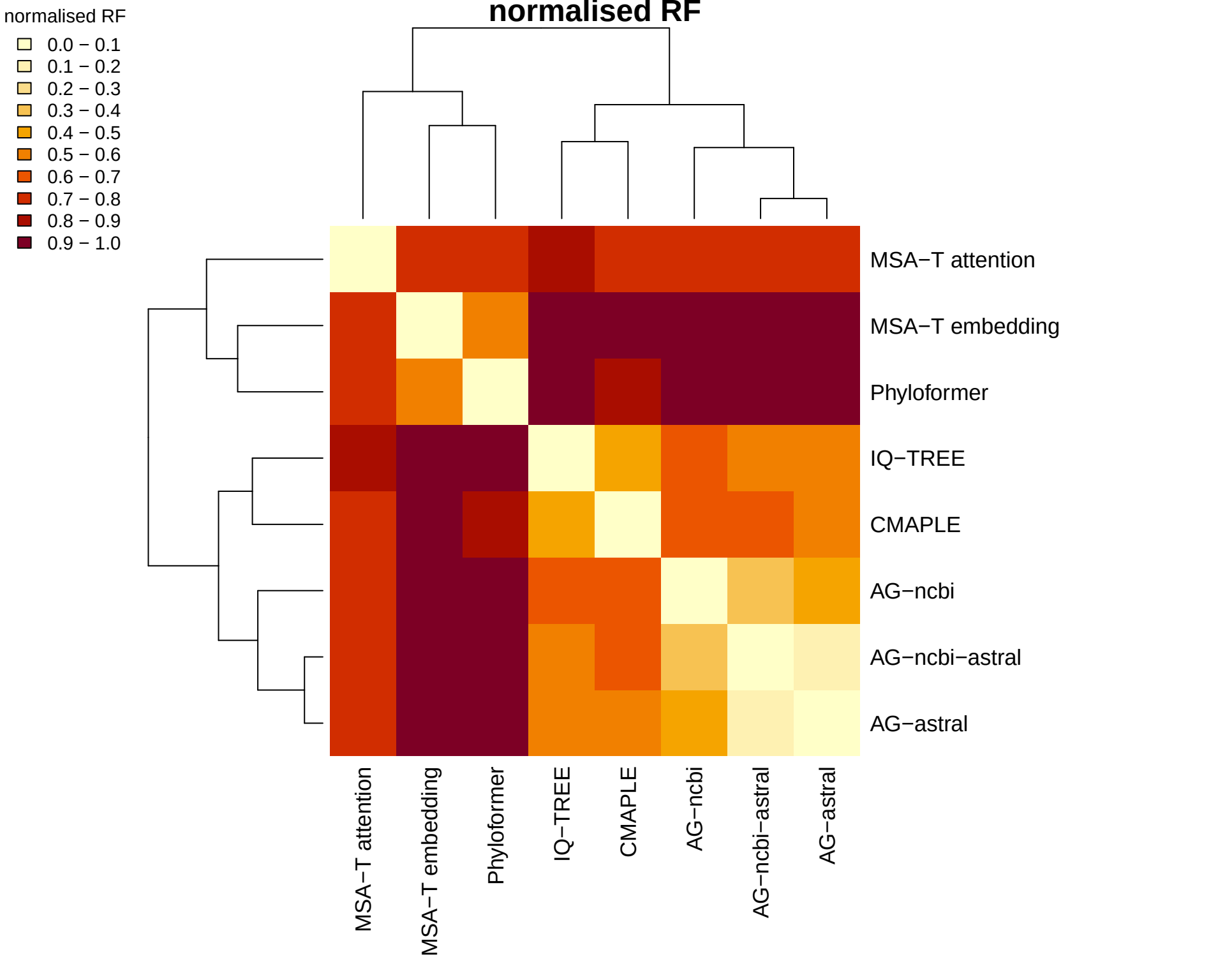
# IQ−TREE vs AG−astral

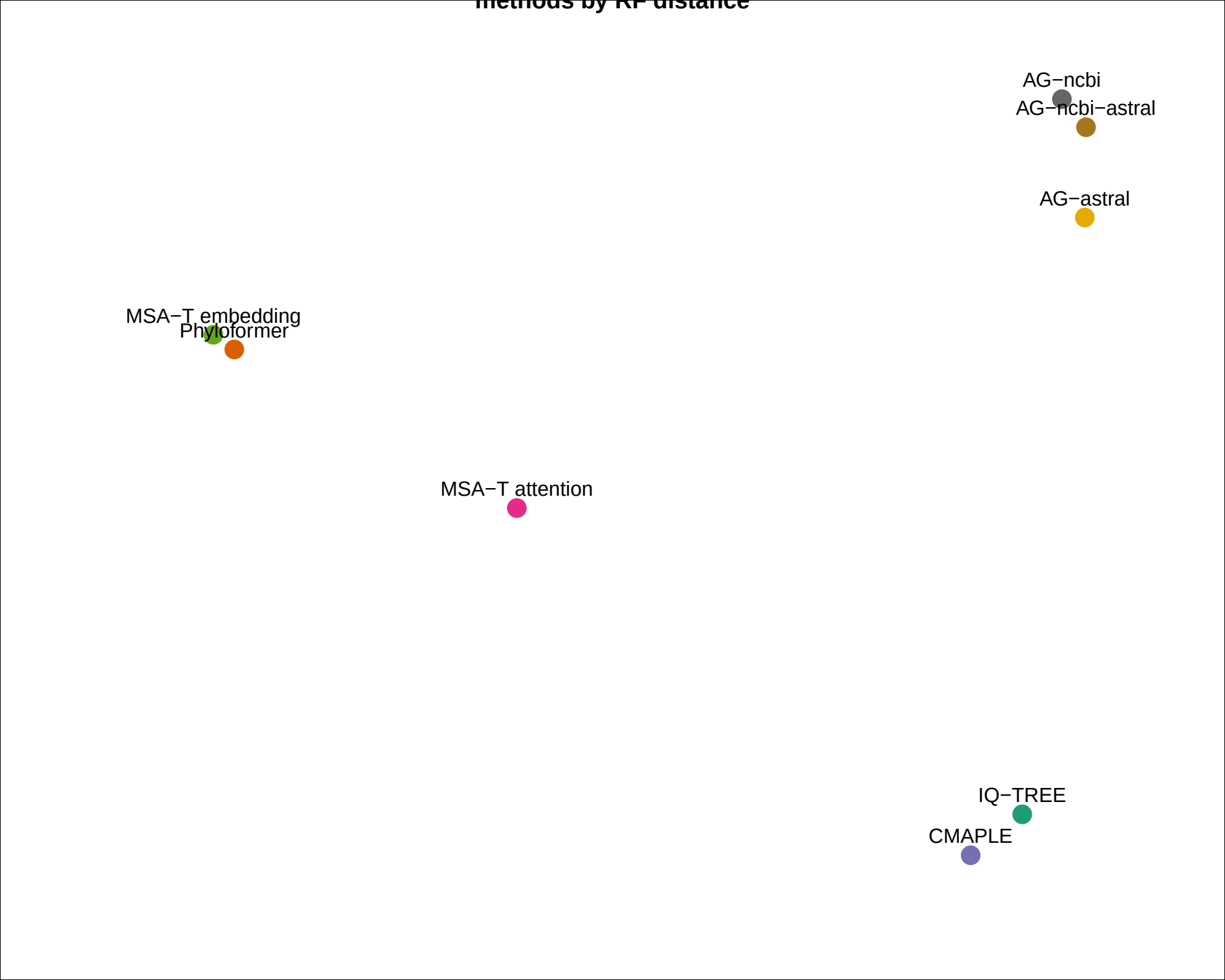


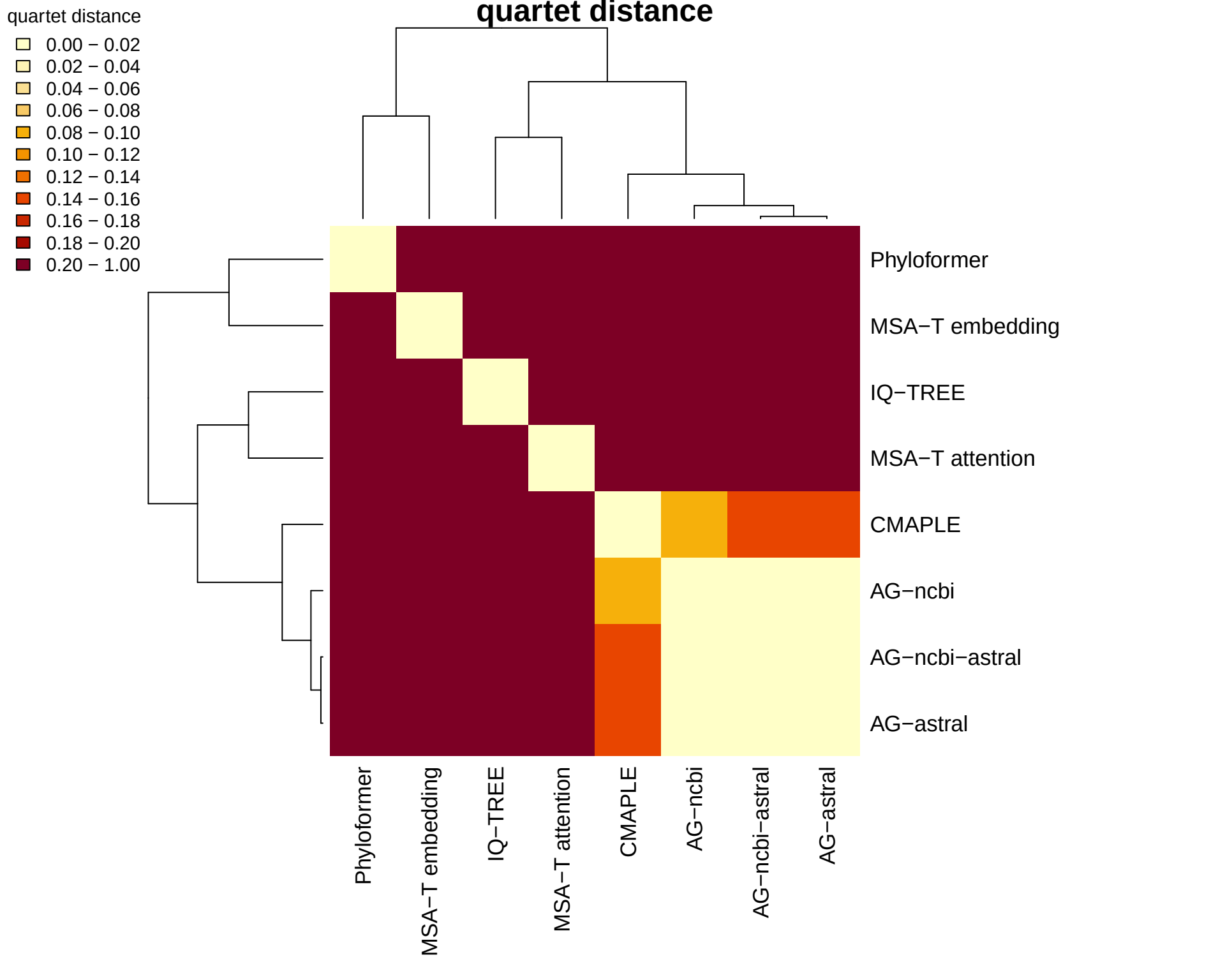
# IQ-TREE vs AG-ncbi-astral

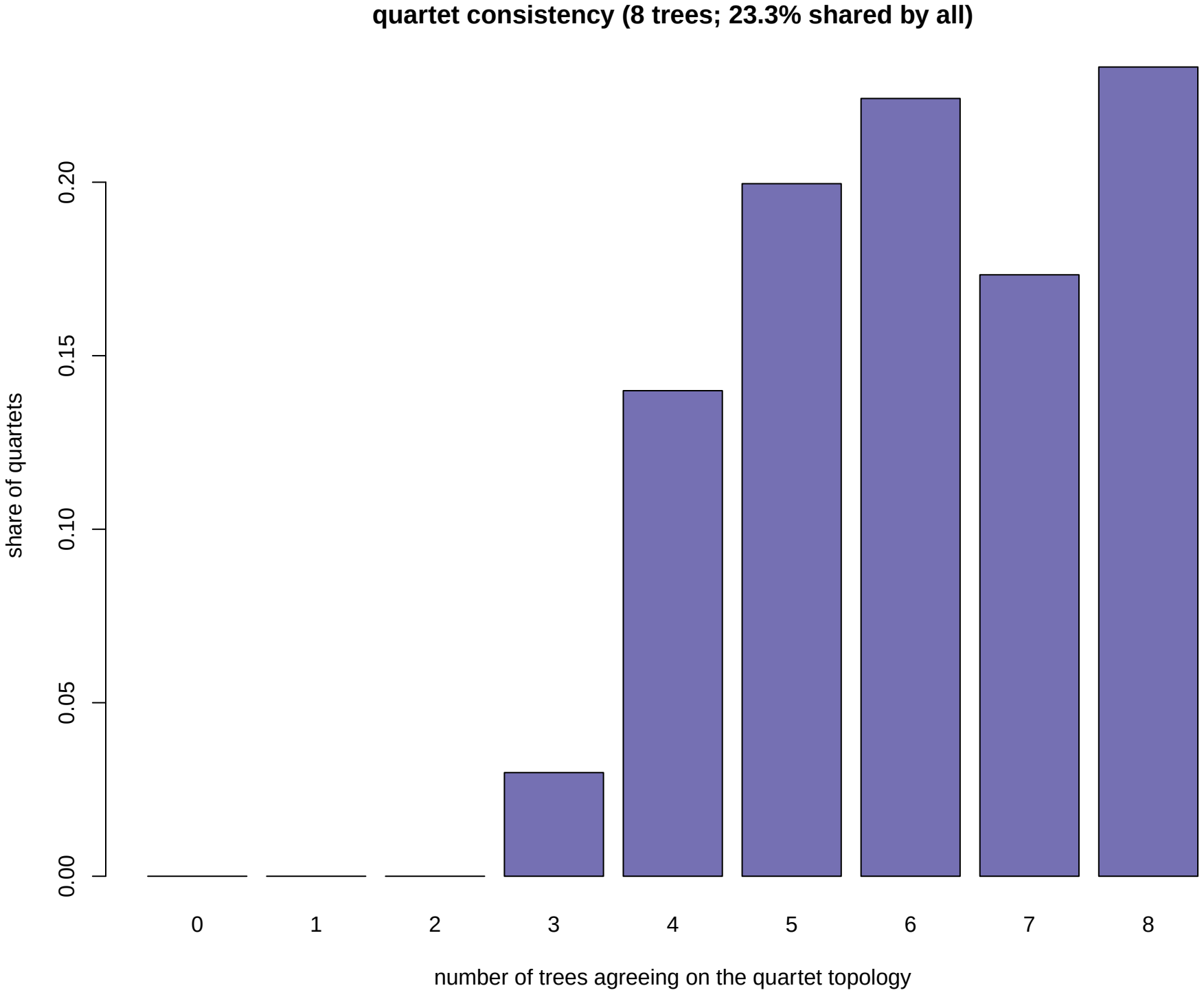


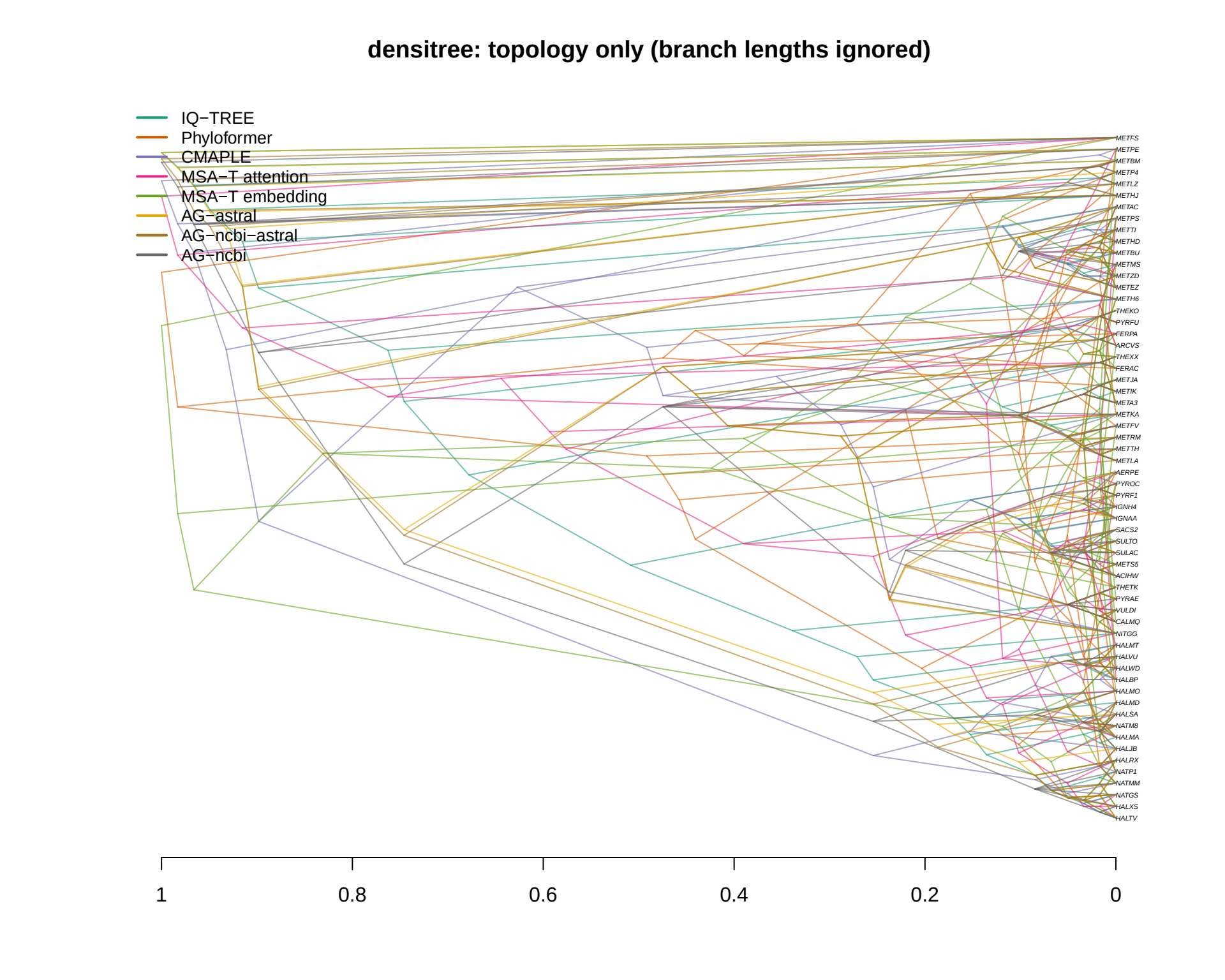

IQ−TREE vs AG−ncbi
METFS
METPE
METBM
METLZ
METP4
METHJ
METAC
METPS
METHD
METTI
METMS
METZD
METEZ
METBU
METH6
THEKO
PYRFU
ARCVS
FERPA
HALMT
HALVU
HALWD
HALBP
NATGS
HALXS
HALTV
NATP1
NATMM
HALRX
HALJB
HALMA
NATM8
HALSA
HALMD
HALMO
NITGG
VULDI
CALMQ
THETK
PYRAE
METS5
ACIHW
SULTO
SULAC
SACS2
IGNAA
IGNH4
AERPE
PYROC
PYRF1
THEXX
FERAC
METKA
METFV
METRM
METLA
METTH
META3
METIK
METJA
METHJ
METLZ
METBM
METP4
METPS
METAC
METMS
METHD
METZD
METEZ
METTI
METBU
METH6
HALVU
HALMT
HALWD
HALBP
NATGS
HALXS
NATP1
NATMM
HALTV
HALRX
HALMO
HALJB
HALSA
HALMA
NATM8
HALMD
METKA
THEXX
FERAC
THEKO
PYRFU
ARCVS
FERPA
NITGG
VULDI
THETK
CALMQ
PYRAE
METS5
ACIHW
SULTO
SULAC
SACS2
AERPE
IGNAA
IGNH4
PYROC
PYRF1
METFV
METRM
METLA
METTH
META3
METIK
METJA

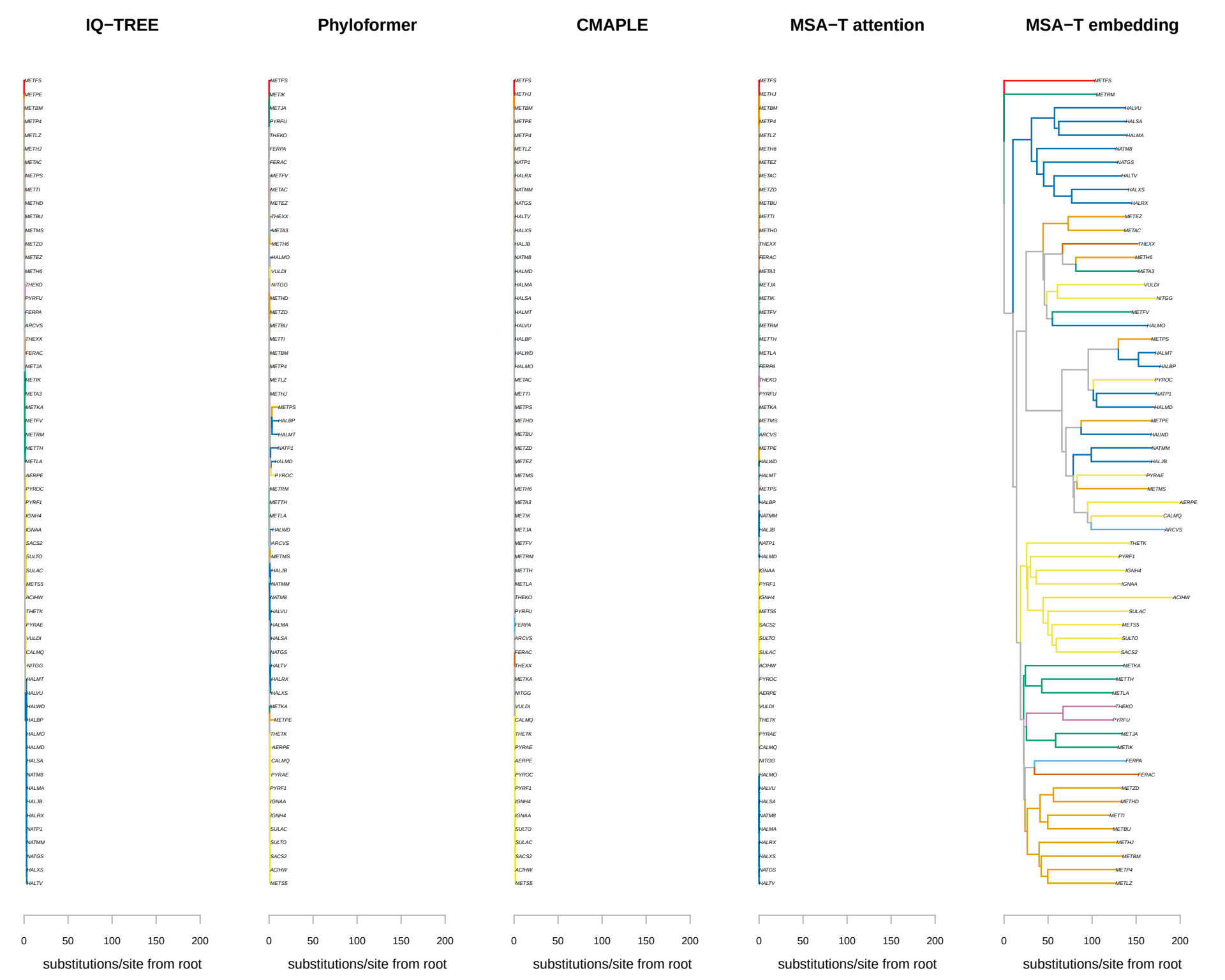

normalised RF
normalised RF
0.0 − 0.1
0.1 − 0.2
0.2 − 0.3
0.3 − 0.4
0.4 − 0.5
0.5 − 0.6
0.6 − 0.7
0.7 − 0.8
0.8 − 0.9
0.9 − 1.0
MSA−T attention
MSA−T embedding
Phyloformer
IQ−TREE
CMAPLE
AG−ncbi
AG−ncbi−astral
AG−astral
MSA−T attention
MSA−T embedding
Phyloformer
IQ−TREE
CMAPLE
AG−ncbi
AG−ncbi−astral
AG−astral


methods by RF distance
AG−ncbi
AG−ncbi−astral
AG−astral
MSA−T embedding
Phyloformer
MSA−T attention
IQ−TREE
CMAPLE

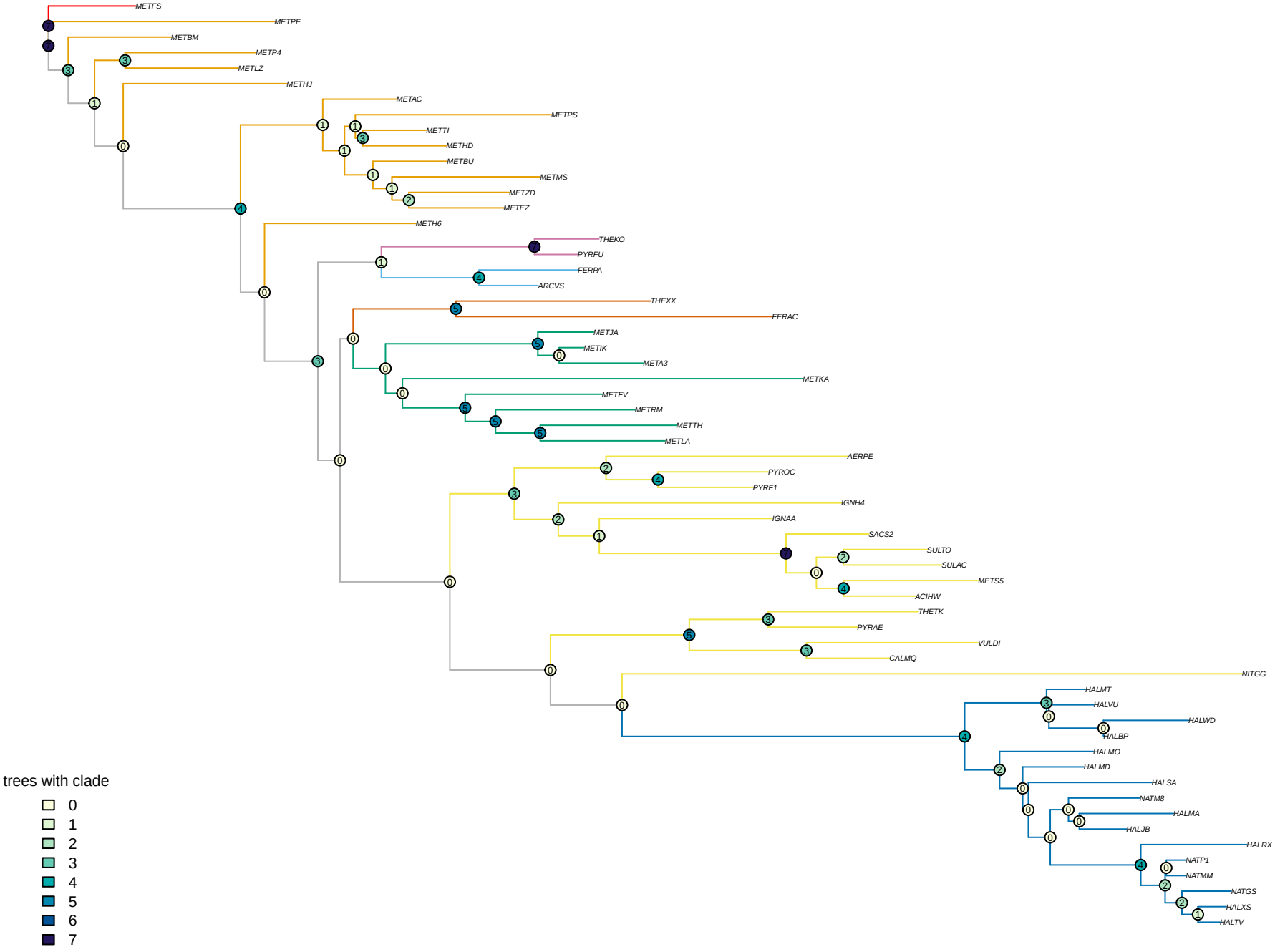

quartet distance
quartet distance
0.00 − 0.02
0.02 − 0.04
0.04 − 0.06
0.06 − 0.08
0.08 − 0.10
0.10 − 0.12
0.12 − 0.14
0.14 − 0.16
0.16 − 0.18
0.18 − 0.20
0.20 − 1.00
Phyloformer
MSA−T embedding
IQ−TREE
MSA−T attention
CMAPLE
AG−ncbi
AG−ncbi−astral
AG−astral
Phyloformer
MSA−T embedding
IQ−TREE
MSA−T attention
CMAPLE
AG−ncbi
AG−ncbi−astral
AG−astral


quartet consistency (8 trees; 23.3% shared by all)
share of quartets
0.00
0.05
0.10
0.15
0.20
0
1
2
3
4
5
6
7
8
number of trees agreeing on the quartet topology

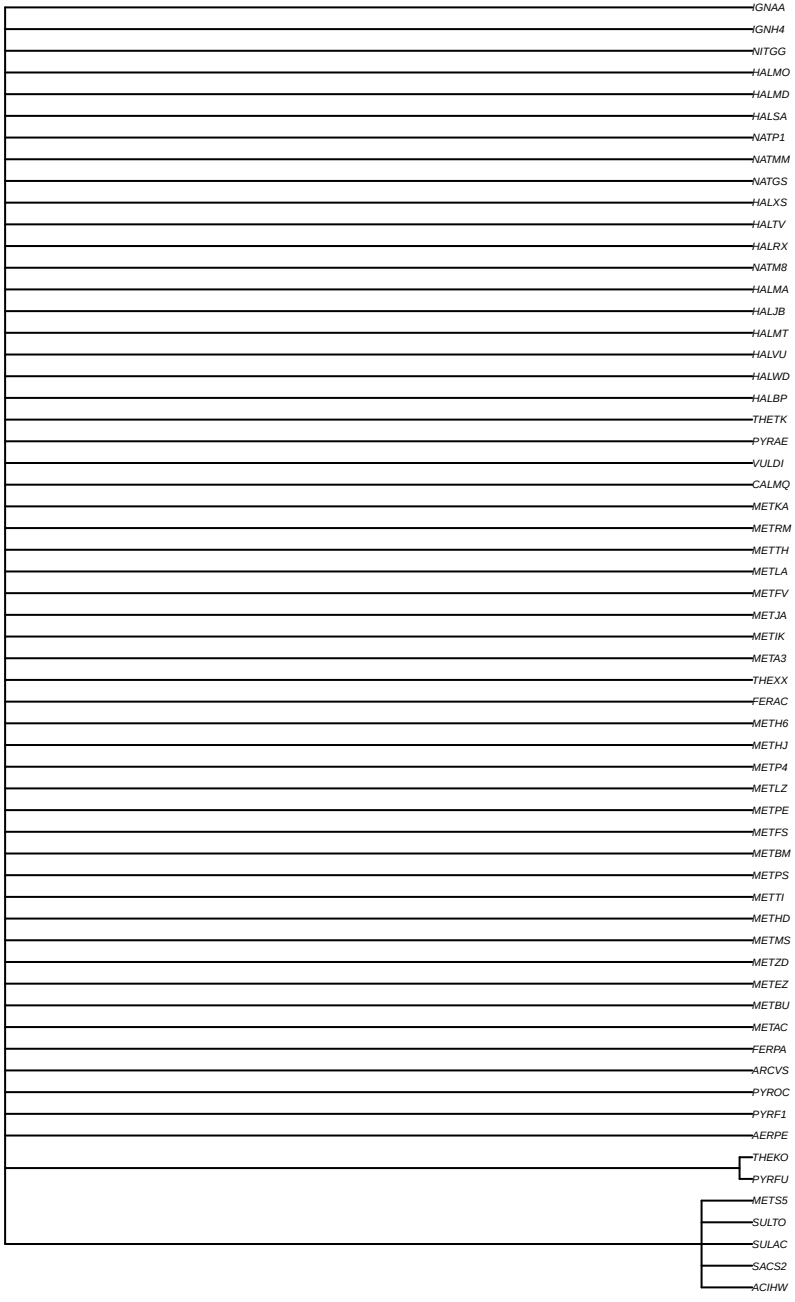

densitree: topology only (branch lengths ignored)
IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
AG−astral
AG−ncbi−astral
AG−ncbi
METFS
METPE
METBM
METP4
METLZ
METHJ
METAC
METPS
METTI
METHD
METBU
METMS
METZD
METEZ
METH6
THEKO
PYRFU
FERPA
ARCVS
THEXX
FERAC
METJA
METIK
META3
METKA
METFV
METRM
METTH
METLA
AERPE
PYROC
PYRF1
IGNH4
IGNAA
SACS2
SULTO
SULAC
METS5
ACIHW
THETK
PYRAE
VULDI
CALMQ
NITGG
HALMT
HALVU
HALWD
HALBP
HALMO
HALMD
HALSA
NATM8
HALMA
HALJB
HALRX
NATP1
NATMM
NATGS
HALXS
HALTV
1
0.8
0.6
0.4
0.2
0


IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
METFS
METPE
METBM
METP4
METLZ
METHJ
METAC
METPS
METTI
METHD
METBU
METMS
METZD
METEZ
METH6
THEKO
PYRFU
FERPA
ARCVS
THEXX
FERAC
METJA
METIK
META3
METKA
METFV
METRM
METTH
METLA
AERPE
PYROC
PYRF1
IGNH4
IGNAA
SACS2
SULTO
SULAC
METS5
ACIHW
THETK
PYRAE
VULDI
CALMQ
NITGG
HALMT
HALVU
HALWD
HALBP
HALMO
HALMD
HALSA
NATM8
HALMA
HALJB
HALRX
NATP1
NATMM
NATGS
HALXS
HALTV
METHJ
METRM
METBM
METPE
METP4
METLZ
METH6
METEZ
METAC
METZD
METBU
METTI
METHD
THEXX
FERAC
META3
METJA
METIK
METFV
METRM
METTH
METLA
FERPA
THEKO
PYRFU
METKA
METMS
ARCVS
METPE
HALWD
HALMT
METPS
HALBP
NATMM
HALJB
NATP1
HALMD
IGNAA
PYRF1
IGNH4
METS5
SACS2
SULTO
SULAC
ACIHW
PYROC
AERPE
VULDI
THETK
PYRAE
CALMQ
NITGG
HALMO
HALVU
HALSA
NATM8
HALMA
HALRX
HALXS
NATGS
HALTV
METFS
METRM
HALVU
HALSA
HALMA
NATM8
NATGS
HALTV
HALXS
HALRX
METEZ
METAC
THEXX
METH6
META3
VULDI
NITGG
METFV
HALMO
METPS
HALMT
HALBP
PYROC
NATP1
HALMD
METPE
HALWD
NATMM
HALJB
PYRAE
METMS
AERPE
CALMQ
ARCVS
THETK
PYRF1
IGNH4
IGNAA
ACIHW
SULAC
METS5
SULTO
SACS2
METKA
METTH
METLA
THEKO
PYRFU
METJA
METIK
FERPA
FERAC
METZD
METHD
METTI
METBU
METHJ
METBM
METP4
METLZ
0
50
100
150
200
substitutions/site from root

# IQ−TREE clades: number of the other 7 trees that contain each clade



## strict




## majority rule

